LESSON PLAN

ADVANCED AIRCRAFT TECHNOLOGY (20)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	07
Total No. of Lesson Plans	20	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Functions and operation of DMM, Spectrum analyzer and function generator.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Functions and operation of DMM, Spectrum analyzer and function generator.
5. Specific Objectives	20.3.1.1 Trainee will be able to explain the construction and operation of DMM & CRO.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of DMM, Spectrum Analyzer, and Function Generator operation; Safety precautions regarding high voltage and probe handling; Instructor supervision during live circuit testing.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a Digital Multimeter?
(b) Define the role of a Function Generator in signal testing.
(c) List three basic components found in a typical DMM.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Digital Multimeter construction and internal circuitry • Operation modes of DMM including voltage and resistance • Safety precautions and probe handling techniques • Common DMM errors and troubleshooting methods	L & Ds	PPT, IB	15 min
(b)	• Function Generator waveforms and frequency control • Spectrum Analyzer signal processing and display • Interconnection of generators and analyzers for testing • Calibration procedures for measurement instruments	L & Ds	PPT, PDF	20 min
(c)	• Advanced DMM features and data logging capabilities • Spectrum Analyzer resolution bandwidth and sweep time • Function Generator modulation types and applications • Maintenance and storage of electronic test equipment	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between RMS and peak voltage measurements on a DMM.
(b) Describe how a Spectrum Analyzer converts time-domain signals to frequency domain.
(c) Identify two safety hazards associated with operating a Function Generator.

12. Home Assignment and Follow Up	Prepare a brief report on the specific applications of DMM and Spectrum Analyzer in aircraft engine monitoring systems, referencing LP 02.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction to CRO and construction details.	Refer to the master syllabus reference list	LP 08

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min